

The data scientist, in depth

Turning governed data into a prediction that survives production

M Usman

A close look at the profession that has to make a model survive production.

Where this is heading

Everyone wants the model. Few can make it ship and stay honest.

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The role everyone wants, and the one most likely to stall

- AI and machine-learning specialists are among the **fastest-growing roles to 2030** (WEF, 2025), and the global AI wage premium has reached **56%** (PwC, 2025).
- In the US, data-scientist employment is projected to grow **34% from 2024 to 2034**, much faster than the average occupation (US BLS, 2024).
- The job has moved past the notebook:
 - Python and statistical rigour are assumed; a model that cannot be evaluated honestly is worth nothing
 - **LLMs, generative AI, and retrieval-augmented generation are the fastest-rising requirement** at the top of the role
 - The decisive question is no longer "can you train a model" but "will the score survive production"

The market is not short of models. It is short of models that ship and stay honest.

The slow step

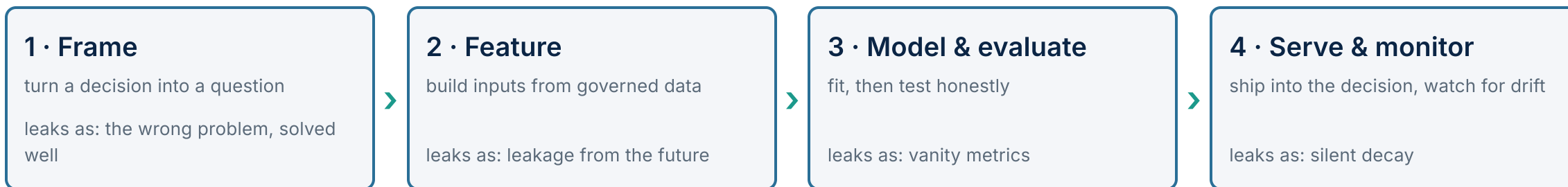
A score that is spectacular offline and fictional in production

The hard part of the role is not fitting a model. A library does that in a line. It is making the number you report the number you will see in production. Information from the future or the target slips into training, the metric looks superb, and the model collapses the day it meets real data.

Treat a result that looks too good as an alarm, not a triumph. The four stages that follow are where a prediction is made honest, or quietly faked.

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The work, end to end



The four stages are one job: a governed dataset turned into a prediction wired to a decision, whose offline score holds up in production. The next four slides take each in turn.

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1 • Frame: turn a decision into a question

A business decision translated into a precise predictive question, with the metric, the baseline, and the cost of a wrong answer agreed before any modelling starts.

Practical steps

- State the decision the model serves and the action its output triggers; if no action changes, do not build it
- Choose the **target and the unit** carefully; a vague label is a model that learns the wrong thing
- Set a real **baseline** first: the current rule, last week's value, or a simple heuristic to beat
- Agree the metric from the decision, and weigh the cost of a false positive against a false negative

The pitfall: the wrong problem, solved well. A technically strong model answers a question nobody needed, or one framed so loosely that its accuracy means nothing.

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2 • Feature: build inputs from governed data

Predictive inputs engineered from the modelled data, with a hard wall between what is known at decision time and what is not.

Practical steps

- Build features only from data available **at the moment of prediction**; anything later is leakage
- Fit every transformation, scaling, encoding, imputation, on the training fold alone, never the full set
- Audit each feature for a path back to the target; a near-perfect predictor is usually a leak
- Where features are shared, use a feature store so training and serving read the same definition

The pitfall: leakage from the future. A column computed after the outcome, or a statistic fitted across the test set, quietly hands the model the answer.

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3 · Model & evaluate: fit, then test honestly

A model chosen for the problem and evaluated under a protocol that mirrors production, so the reported score is the score it will earn.

Practical steps

- Split train, validation, and test **first**; match the protocol to the data with stratified, grouped, or time-ordered folds
- Start simple: a strong baseline often beats a complex model that cannot be explained or trusted
- Choose the metric from the decision, precision and recall, F1, or PR-AUC, not raw accuracy on imbalanced data
- Resample **inside** each fold, keep an untouched test set, and check **calibration**, not only ranking

The pitfall: vanity metrics. A headline accuracy on a leaked or imbalanced split that says nothing about the decision the model will drive.

Illustrative tools: scikit-learn, XGBoost, PyTorch, TensorFlow · MLflow for tracking · SHAP for explanation.

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4 · Serve & monitor: ship into the decision, watch for drift

The model packaged behind a versioned interface, deployed into the decision path, and watched, so silent decay is caught before it costs anything. Here data science meets MLOps.

Practical steps

- Package model, features, and dependencies together; verify the serving score matches the training score
- Release progressively, shadow or canary, with a fast rollback path
- Monitor input **data drift**, **concept drift**, and **ground-truth performance**, not only uptime
- Trigger retraining on a measured drop or a drift threshold, not the calendar alone

The pitfall: silent decay. A model that ships without monitoring degrades as the world moves, found out only when a business metric slips long after the cause.

Illustrative tools: FastAPI, BentoML, KServe · Docker · Evidently, NannyML · a model registry.

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The specialisms, and the rising edge

- **Time-series and forecasting.** Demand, load, and failure prediction, where the split must respect time or the evaluation lies.
- **Graph machine learning.** When the relationships carry the signal, entity resolution, fraud rings, network effects, graph neural networks read structure a table cannot.
- **Natural language and text.** From classification to retrieval over a corpus; the foundation under most enterprise LLM work.
- **LLMs, generative AI, and RAG.** The fastest-rising edge: retrieval-augmented generation, evaluation of generated output, and the discipline to keep it grounded.

A scientist is hired on the floor, Python and honest evaluation, and chosen on the specialism. The edge today is a real, evaluated LLM or RAG project.

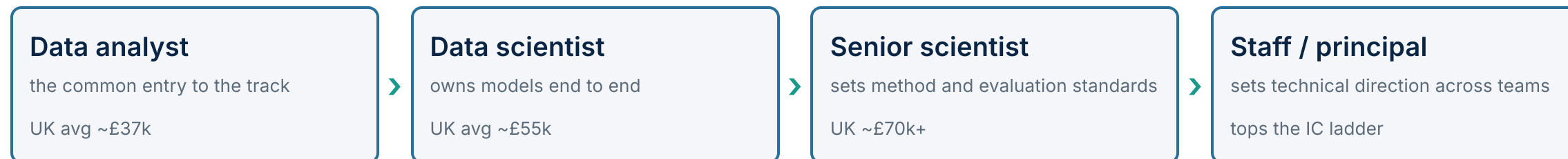
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How a data scientist is measured

- **The decision metric, not the algorithm.** Precision and recall, or PR-AUC, tied to the action; raw accuracy is rarely the right answer.
- **Lift over a real baseline.** Whether the model beats the current rule by enough to justify its cost and complexity.
- **Production parity.** Whether the offline score holds when the model meets live data, and stays calibrated under drift.
- **Whether it ships and survives.** A brilliant notebook that never reaches a decision has produced nothing.

The discipline is not the model. It is the honesty of the evaluation around it.

The ladder: from analyst to principal



For many employers the analyst seat is the entry to the science track, with the scientist a step above it. Seniority can cross the lines. An experienced analyst may out-earn a newly hired scientist, but the median of each role keeps analyst below scientist. UK role averages, Glassdoor 2025/26; elite tech and finance pay well above these.

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The roadmap a data scientist climbs

5 · The frontier and production

LLMs, RAG, and generative AI; MLOps, serving, and monitoring; the served, watched model

4 · A specialism

time-series, graph ML, or NLP in depth; the edge that gets you hired over the median

3 · Modelling and honest evaluation

ML fundamentals; leakage-safe protocols; the metric chosen from the decision

2 · Data and SQL

databases and SQL; the shape, grain, and meaning of the data

1 · Foundations

Python, version control, and the mathematics and statistics under everything above

The first two rungs are shared with every data role; the scientist's signature is honest evaluation, then a specialism, then the production frontier at the top.

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What good looks like

- **Start from the decision, end at the decision.** A model earns its place only if it changes an action and beats the baseline that action already uses.
- **Treat a brilliant score as a suspect.** The first question of an excellent result is "where did it leak", not "where do we deploy".
- **Evaluate like production, not like a leaderboard.** Time-ordered or grouped splits, calibration, and an untouched test set, fitted per fold.
- **Lead with a real, evaluated project.** A served model with honest numbers beats a notebook of high accuracies every time.

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The lens behind this view

My view of this stage is the work of the doctorate. My research is on efficient, proactive AI, where honest evaluation and the cost of a wrong prediction are the problem itself. It spans graph attention networks for failure-aware scheduling and time-series failure prediction. The baseline, the leakage audit, and the metric chosen from the decision are the discipline of that research, and of this deck.

The work is public and reproducible. Four worked examples on my profile carry their numbers in full, the flattering results and the ones that need explaining. They cover a served sentiment service with a retrieval-augmented layer, a time-series forecaster benchmarked against naïve baselines, and two clinical-risk studies with their leakage analysis stated openly. I also teach the data-mining foundations the role is built on.

The research and the four worked examples are public: mtusman-ai.github.io/M.Usman

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In one line

The data scientist turns governed data into a prediction wired to a decision. Frame the question, build leakage-free features, evaluate as honestly as production will, and ship behind a monitored interface. The skill is not the model; it is the honesty of the evaluation around it.

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Sources

WEF Future of Jobs (2025): AI and machine-learning specialists among the fastest-growing roles to 2030 · PwC Global AI Jobs Barometer (2025): 56% global AI wage premium · US BLS Occupational Outlook Handbook (2024): data-scientist employment projected to grow 34% from 2024 to 2034 · Role-average UK salaries: Glassdoor UK (2025/26). The evaluation practice draws on the leakage, cross-validation, and MLOps literature: Google Rules of ML and the ML Test Score; the scikit-learn and MLflow documentation; the drift-monitoring references. The per-stage learning path follows the standard curriculum progression: foundations, then SQL and databases, then modelling and evaluation, then a specialism, then LLM workflows and MLOps. Tools named are illustrative, not endorsements.

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See the full portfolio



The work behind this deck sits on one page: the Insights series, the projects and the code that runs them, and the publications.

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